

Education Background

University of Southern California (USC), USA	09/2021 – 05/2023
Master of Engineering in Electric Computer and Engineering	GPA: 3.1/4.0
Changchun University of Science and Technology (CUST), China	09/2017 – 05/2021
Bachelor of Science Electric Computer and Engineering	GPA: 3.83/4.0

Selected Published Paper

- **Membership Inference Attacks on In-Context Learning Recommendation**, ACM RecSys 2026
- **Auditing approximate machine unlearning for differentially private models**. IEEE ICDM 2025
- **RecPS: Privacy Risk Scoring for Recommender System**. ACM RecSys 2025
- **Adaptive Domain Inference Attack with Concept Hierarchy**. KDD 2025
- **FT-PrivacyScore: Personalized Privacy Scoring Service for Machine Learning Participation**. ACM CCS 2024

Work Experience

Hithink Researcher (AInvest)	New York, NY
AI Algorithm Researcher (AIME)	08/2026 - present
• Designed a schema-driven KYC user modeling framework and automated feature extraction pipeline that integrates heterogeneous signals from trading history, portfolio holdings, financial Q&A, news consumption, and market interactions; developed a unified representation layer to migrate legacy user profiles into an extensible KYC feature taxonomy with continual profile updates. • Architected a multi-agent framework for longitudinal financial user modeling, decomposing profile updates into candidate evidence extraction, relevance assessment, profile refinement, and memory writing; introduced short-, medium-, and long-term hierarchical memory with gated update policies to control when user information is retained, reinforced, deferred, or revised. • Developed mechanisms for evidence-aware profile evolution, allowing agents to distinguish transient behavioral signals from persistent user characteristics and maintain temporally consistent investor representations across multi-source behavioral sequences.	
Marquette University	Milwaukee, WI
AI Researcher	04/2024 - 08/2024
• Designed and implemented FT-PrivacyScore, a privacy evaluation framework for quantifying sample-level privacy risks in LLM fine-tuning, providing interpretable risk estimates under controlled data-access settings while preserving downstream model utility. • Built an end-to-end LLM supervised fine-tuning (SFT) pipeline for privacy-risk analysis, including sensitive-domain data preprocessing, instruction construction, supervised fine-tuning, and LoRA-based parameter-efficient adaptation; systematically evaluated how individual training samples affect privacy leakage under different data-participation settings. • Developed and deployed a session-aware LLM question recommendation system for an online course platform, modeling dynamic user intent from multi-turn query histories and generating progressive follow-up questions with topic-continuity modeling, diversity control, and semantic deduplication; increased recommendation coverage from 55% to 100% and user conversion from 20% to 40%.	
CodersData	Los Angeles, CA
Machine Learning Engineering Intern	08/2023 - 01/2024
• Developed an LLM-powered proactive personalization system that jointly modeled user profiles, historical queries, and behavioral sequences to capture evolving interests; designed context-aware generation, multi-channel retrieval, and ranking strategies and iteratively optimized the system through online A/B testing. • Built large-scale recommendation data pipelines with Databricks and Apache Spark, processing impression, click, and conversion logs for sample construction, feature aggregation, and training-data generation, supporting iterative CTR modeling and online experimentation. • Designed and analyzed A/B experiments for recommendation and product strategies, optimizing content recommendation and interaction flows around user pain points and improving subscription conversion by 3.7%. • Collaborated with product, growth, and content teams to define acquisition, retention, and engagement metrics and translate behavioral insights into recommendation strategies, increasing TikTok outbound video views by 5.7% and CTR by 1.2%.	
Hirebeat.Inc	Jersey City, NJ
Data Scientist Intern	01/2023 – 05/2023
• Built an NLP-based résumé parsing pipeline and user résumé data lake on AWS S3, extracting structured attributes including skills, work experience, and job preferences from uploaded résumés; improved core field extraction accuracy by 3.7%. • Integrated extracted representations into a résumé-job recommendation system, improving candidate-job matching through feature engineering, model optimization, and controlled experiments across multiple feature configurations. • Designed and executed A/B tests combining user-behavior and résumé features to quantify their contributions to recommendation quality and downstream conversion, increasing overall customer conversion by approximately 1.5%. • Developed end-to-end data processing and analytics workflows using advanced SQL including CTEs and stored	

procedures for near-real-time ETL, and integrated results with **AWS QuickSight** for monitoring and analysis.

Research Experience

University of Maryland, Baltimore County

08/2024 - present

Research Assistant

Baltimore, MD

Project 1: Adaptive Domain Inference Attack with Concept Hierarchy (KDD 2025 & ICDM 2025)

- Developed **efficient ML auditing methods** for deployed model APIs and machine unlearning systems, including **ADI** for black-box domain inference and **A-LiRA** for auditing approximate unlearning.
- Built scalable evaluation pipelines using concept hierarchy search, data augmentation, LiRA-style likelihood testing, shadow models, DP-SGD, and FFCV-accelerated training to analyze model behavior under limited access and compute budgets.
- Evaluated ResNet-18/ResNet-152 models using **AUC**, **TPR@1%FPR**, OTDD, and privacy-risk metrics; reduced LiRA auditing cost by **88.3%** while maintaining comparable evaluation quality.

Project 2: Trustworthy Recommendation (published on RecSys 2025)

- Developed **RecPS**, a large-scale recommender-system evaluation framework that analyzes user–item interaction data and produces fine-grained privacy-risk scores at both interaction and user levels.
- Built **RecLiRA**, a LiRA-based interaction-level model analysis framework for recommender systems, designed to evaluate model families including **neural collaborative filtering**, **graph-based collaborative filtering**, and ranking-oriented RecSys architectures using target/shadow model training.
- Designed scalable RecSys experiments across **millions of user–item interactions** using chronological train/validation/test splits, negative sampling, **HR@K**, AUC, and low-FPR evaluation; accelerated repeated target/shadow model training with **multi-GPU infrastructure**.
- Demonstrated that score-guided removal of sensitive interactions preserves recommendation utility better than full user-level removal, supporting **responsible AI**, privacy-aware recommender development, and data-removal decision workflows.
- Extended **RecEraser**, **UltraRE**, to study **recommendation unlearning** beyond privacy, **including removal of dirty interactions, outdated user preferences, and low-quality behavioral signals while preserving useful long-term and short-term user interest patterns**.

Project 3: LLM-based Recommendation (RecSys 26)

- Developed LLM-powered recommendation pipelines using in-context learning, knowledge-graph retrieval, prompt-based adaptation, and conversational explanation generation for personalized recommendation and user-preference modeling.
- Built **graph-aware RAG workflows** that retrieve, re-rank, and encode structured knowledge subgraphs to ground LLM recommendations, improving factuality and reducing hallucination caused by outdated or missing domain knowledge.
- Evaluated **credible and persuasive recommendation explanations** using LLM-based and human-aligned scoring protocols, analyzing whether generated explanations remain faithful to item facts while improving user acceptance.
- Designed privacy and robustness audits for **ICL-based LLM RecSys**, including similarity, memorization, inquiry, and poisoning-style membership inference attacks against prompt-based user interaction examples.

University of Science and Technology of China

Hefei, China

Undergraduate Researcher (Mentor: Bensheng Qiu)

12/2019 – 01/2021

- Studied **representation learning for high-dimensional spatiotemporal data**, using task-based functional MRI (tfMRI) as a real-world testbed and designing features to capture informative spatial and temporal patterns.
- Designed and implemented a **3D CNN-based encoder–decoder architecture** incorporating temporal embeddings and channel-wise attention for compact representation learning and task-state prediction.
- Extended the framework with a **3D Swin Transformer backbone** and conducted controlled comparisons to study how convolutional locality and window-based self-attention behave under high-dimensional and limited-data regimes.
- Evaluated the proposed models on two **Human Connectome Project (HCP)** benchmarks, achieving **94.0 ± 8.2%** accuracy on motor-task decoding and **79.6 ± 7.1%** on gambling-task decoding, demonstrating robust performance across heterogeneous prediction tasks.

In Progress

- **Secure-by-Disguise: A Systematic Evaluation of Image Disguising for Confidential Medical Image Modeling**
- **Do LLM simulators learn personalized conditional behavior, or do they collapse heterogeneous investors into a generic average investor?**
- **Volume-Wise Task fMRI Decoding with Deep Learning: Enhancing Temporal Resolution and Cognitive Function Analysis**
- **Privacy\textit{Déjà Vu}** Effect: Resurfacing Sensitive Samples in Continual Fine-tuning

Skills

- **Programming & ML:** Python, SQL, PyTorch, Hugging Face Transformers, pandas
- **Machine Learning:** CTR Prediction, Learning-to-Rank, Recommendation Systems, Candidate Generation, Retrieval, Re-ranking, Collaborative Filtering, Graph-based Recommendation, A/B Testing
- **LLMs & RAG:** In-Context Learning, RAG, KG-RAG, Prompt Engineering, LoRA/PEFT, LangChain, LlamaIndex, LLM Agents, Conversational Recommendation
- **Data & Cloud:** AWS Glue, Athena, QuickSight, Databricks, MongoDB, ETL Pipelines, Tableau, MongoDB, ElasticSearch

